

Review

Moonlighting Motors: Kinesin, Dynein, and Cell Polarity

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In addition to their well-known role in transporting cargoes in the cytoplasm, microtubule motors organize their own tracks – the microtubules. While this function is mostly studied in the context of cell division, it is essential for microtubule organization and generation of cell polarity in interphase cells. Kinesin-1, the most abundant microtubule motor, plays a role in the initial formation of neurites. This review describes the mechanism of kinesin-1-driven microtubule sliding and discusses its biological significance in neurons. Recent studies describing the interplay between kinesin-1 and cytoplasmic dynein in the translocation of microtubules are discussed. In addition, we evaluate recent work exploring the developmental regulation of microtubule sliding during axonal outgrowth and regeneration. Collectively, the discussed works suggest that sliding of interphase microtubules by motors is a novel force-generating mechanism that reorganizes the cytoskeleton and drives shape change and polarization.

Kinesin-1-Driven Microtubule Sliding in Neurons and Non-Neuronal Cells

The standard and well-studied function of conventional kinesin (also known as kinesin-1 or KIF5) is to transport cargoes along microtubules towards plus-ends. Recently, several studies, including the work of our group, have uncovered a novel 'unconventional' function of kinesin-1 in sliding cytoplasmic microtubules against each other. Biochemical studies demonstrated that, in addition to the well-characterized N-terminal motor domain [1,2], kinesin-1 has another microtubule-binding site at the C terminus of the heavy chain (kinesin-1 heavy chain, KHC) [3,4]. This C-terminal microtubule-binding site contains multiple basic amino acids that are highly conserved across many species (Figure 1). This site can bind to an acidic E-hook at the tubulin C terminus through complementary electrostatic interactions [3,4], and, unlike the N-terminal motor domain, is ATP-independent. Using the N-terminal motor domain and the C-terminal binding site, kinesin-1 can cross-bridge two microtubules and move them against each other (microtubule sliding, Figure 1). In this case, the microtubule that is bound to the C terminus of the heavy chain becomes a 'cargo' for kinesin-1, while the other microtubule serves as a track (Figure 1). When kinesin-1 motor walks on one microtubule (track), it moves another microtubule (cargo) relative to the track microtubule (Figure 1). Experiments overexpressing the kinesin-1 C-terminal tail in tissue culture cells showed that the C-terminal tail decorates microtubules uniformly [5,6], indicating that all microtubules are identical in their ability to bind to the kinesin-1 C-terminal tail. Thus kinesin-1 probably binds to two microtubules in a random orientation (statistically 50% upright and 50% upside-down, simplified as two opposite kinesin-1 motors in Figure 2). Therefore, it is likely that every microtubule serves both as a cargo and as a track at the same time. In this scenario, kinesin-1 only slides antiparallel microtubules against each other (Figure 2A). If it cross-bridges parallel microtubules, the forces of kinesins in opposite orientations are subtracted, and no movement can be produced (Figure 2B).

Trends

Contrary to the textbook view, cytoplasmic microtubules in interphase cells are not static; they robustly slide in the cytoplasm moved by microtubule motors.

Two motors contribute to microtubule sliding: kinesin-1 that slides microtubules against each other, and cytoplasmic dynein that drives microtubule movement relative to the cell cortex.

Sliding microtubules can drive cell shape change by pushing against the plasma membrane.

Plus-end motor kinesin-1 and minus-end motor dynein cooperate to achieve the correct microtubule organization in the axons.

Microtubule sliding activity is developmentally regulated independently of the global regulation of the motor activity.

F-actin in the growth cone of growing neurites antagonizes the neurite outgrowth by preventing sliding microtubules from penetrating into the growth cone.

Cargoes can be transported by hitchhiking on moving microtubules.

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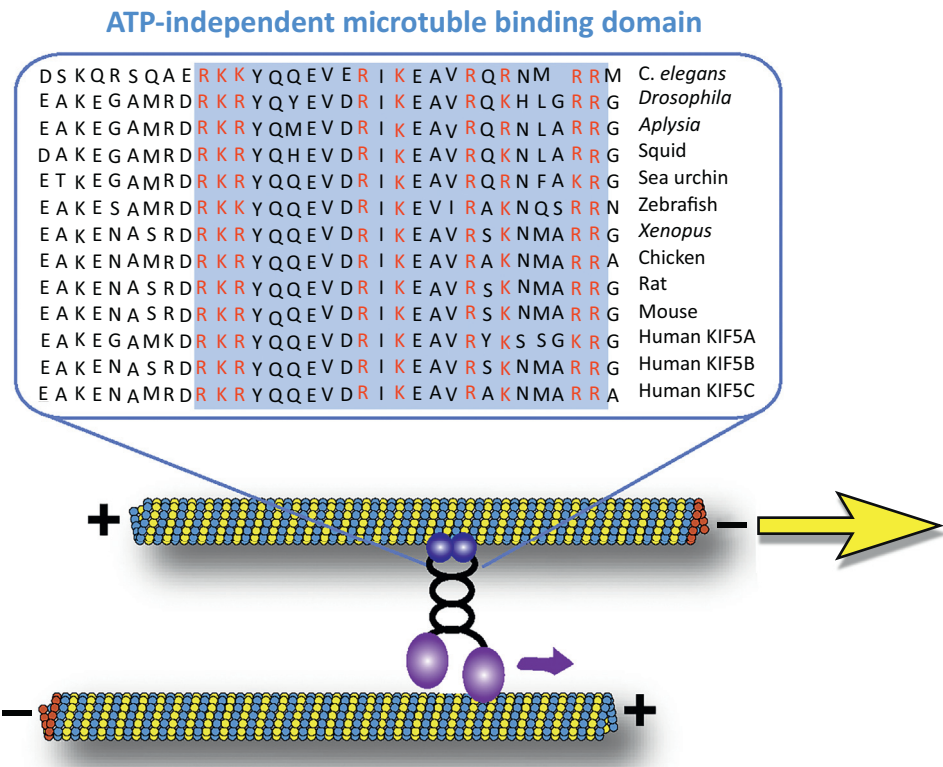


Figure 1. Highly Conserved C-Terminal Microtubule Binding Site of Kinesin-1 Heavy Chain. Amino acids of kinesin-1 heavy chain C-terminal region from 11 species are aligned in Clustal Omega (www.ebi.ac.uk/Tools/msa/clustalo) using the following kinesin heavy chain protein sequences: *Caenorhabditis elegans* (roundworm); *Drosophila melanogaster* (fruit fly), *Aplysia californica* (California sea hare), *Doryteuthis pealeii* (longfin inshore squid), *Strongylocentrotus purpuratus* (purple sea urchin), *Danio rerio* (zebrafish), *Xenopus tropicalis* (western clawed frog), *Gallus gallus domesticus* (chicken), *Rattus norvegicus* (brown rat), *Mus musculus* (house mouse), and *Homo sapiens* (KIF5A, KIF5B, and KIF5C). The ATP-independent microtubule binding site is highlighted in the light blue box, and all basic residues in the microtubule binding site are labeled in red.

Microtubule sliding by kinesin-1 was first proposed based on *in vivo* observations in a study of the fungus *Ustilago maydis*. Straube and colleagues demonstrated that kinesin-1 causes microtubule bending and bundling [7]. To directly visualize microtubule sliding, our group fused a photoconvertible protein tag to tubulin and imaged microtubule movement in

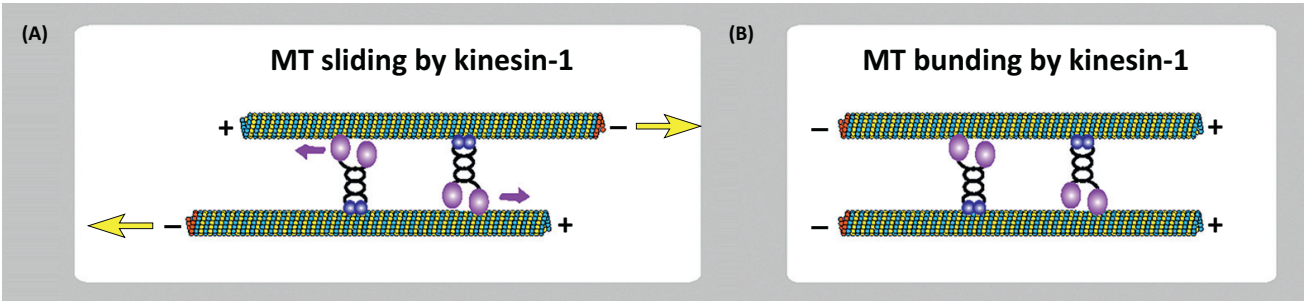
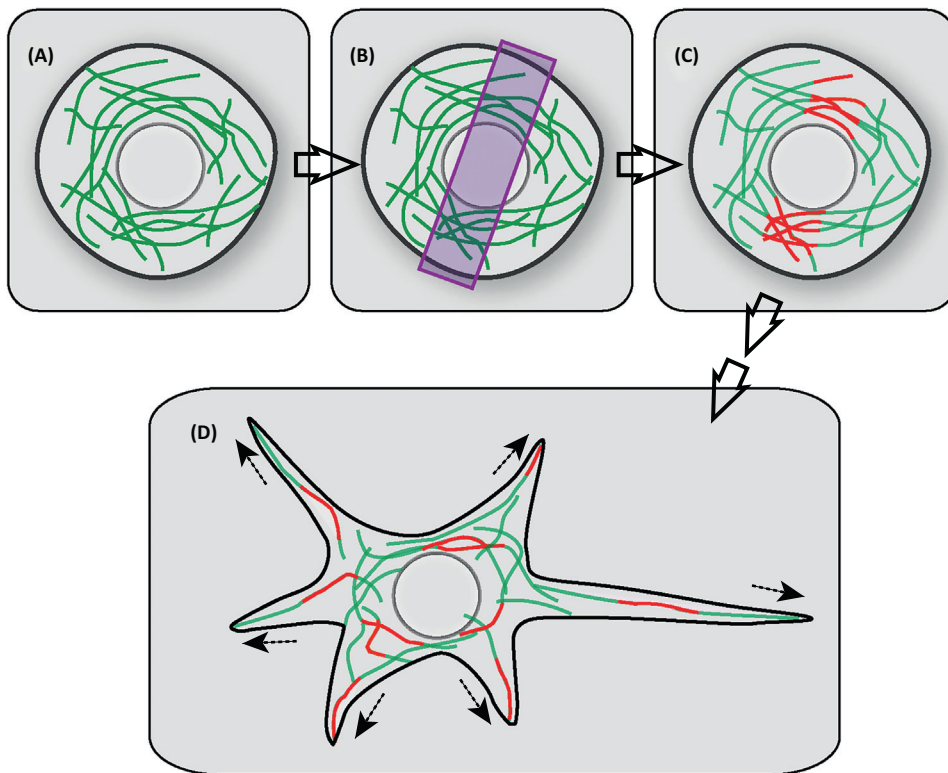


Figure 2. Microtubule Sliding by Kinesin-1 Requires Antiparallel Orientation. Kinesin-1 binding to two microtubules using its motor domain and C-terminal microtubule binding site leads to sliding of antiparallel microtubules (A) and bundling of parallel microtubules (B).



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Figure 3. Visualization of Microtubule Sliding in *Drosophila* Neurons Using Photoconvertible Tubulin. (A) A spherical-shaped young neuron expressing photoconvertible Eos-tagged α -tubulin. (B) 400 nm light is applied to a restricted area to photoconvert a subset of microtubules in the young neurons. (C) A subset of microtubules is photoconverted from green to red. (D) Red microtubule fragments are scattered throughout the cell body and in the newly formed neurites by microtubule–microtubule sliding, revealed by time-lapse movies.

interphase cells. Local photoconversion of this probe can be used to apply fiduciary marks on microtubules. Following these marks (photoconverted microtubule segments) allows visualization of microtubule movements in live cells (Figure 3). We documented robust microtubule sliding in many types of tissue culture cells, including *Drosophila* S2 cells, *Xenopus* fibroblasts, and rat kangaroo epithelial Ptk2 cells [8,9]. Table 1 provides an overview of different approaches to visualizing microtubule sliding.

Experiments in *Drosophila* S2 cells demonstrated that depolymerization of actin cytoskeleton with latrunculin B, or its fragmentation by cytochalasin D before plating, results in formation of multiple radial processes filled with microtubules [8,10–14]. We further demonstrated that this dramatic process formation is driven by kinesin-powered microtubule sliding [8].

We later detected microtubule sliding by kinesin-1 in cultured *Drosophila* primary neurons, including embryonic neurons [15] and larval motor neurons [16], as well as in sensory neurons *in vivo* [6]. Furthermore, not only do microtubules slide in differentiating neurons, but, remarkably, their sliding drives initial extension of neurites in cultured neurons by pushing against the plasma membrane at the tips of the neurites [15–17], consistent with the data that destabilization of actin filaments induces fast axon outgrowth [18]. These data demonstrate that, at least at the initial stages of neurite outgrowth, microtubule sliding by kinesin-1 provides the driving forces for neurite extension [15]. These data are consistent with the previous work in cultured rat

Table 1. Summary of Visualization of Microtubule Sliding and Related Motors

Species	Cell type	Visualization method	Motors involved	Refs
<i>Ustilago maydis</i>	Single cell	GFP- α -tubulin	Kinesin-1 and dynein	Straube <i>et al.</i> 2006 [7]
<i>Drosophila melanogaster</i>	S2 cells (tissue culture)	mCherry-tubulin and Dendra2- α -tubulin (photoconversion);	Kinesin-1	Jolly <i>et al.</i> 2010 [8]
<i>Xenopus</i>	Fibroblast (tissue culture)	Dendra2- <i>Xenopus</i> α -tubulin (photoconversion)	Kinesin-1	Jolly <i>et al.</i> 2010 [8]
Rat	Kangaroo epithelial Ptk2 cells (tissue culture)	GFP-tubulin	Kinesin-1	Jolly <i>et al.</i> 2010 [8]
<i>Drosophila melanogaster</i>	Embryonic neurons (primary culture)	tdEos2- α -tubulin (photoconversion)	Kinesin-1	Lu <i>et al.</i> 2013 [15]
<i>Drosophila melanogaster</i>	Larval motor neurons (primary culture)	tdEos2- α -tubulin (photoconversion)	Kinesin-1	Lu <i>et al.</i> 2015 [16]
<i>Drosophila melanogaster</i>	Sensory neurons underneath the pupal bristles <i>in vivo</i>	tdMaple3- α -tubulin (photoconversion)	Kinesin-1	Winding <i>et al.</i> 2016 [6]
<i>Caenorhabditis elegans</i>	Motor neurons <i>in vivo</i>	EBP-2-GFP (plus-end microtubule marker)	Kinesin-1	Yan <i>et al.</i> 2013 [21]
Mouse	Myoblast C2C12 (tissue culture)	mEos2-tubulin (photoconversion)	Kinesin-1 and dynein	Mogessie <i>et al.</i> 2015 [39]
Amoeba <i>Reticulomyxa</i>	Single cell	Differential interference contrast (DIC) microscopy movie	Dynein	Koonce <i>et al.</i> 1987 [65]; Euteneuer <i>et al.</i> 1989 [66]
Rat	Sympathetic neurons (primary culture)	EGFP-tubulin and photobleaching	Dynein	He <i>et al.</i> 2005 [25]
Mouse/monkey	Neuro-2a and Cos7 (tissue culture)	GFP-tagged and mCherry-tagged microtubule-associated protein MAP2c (fluorescence speckle and total internal reflection fluorescence, TIRF, microscopy)	Dynein	Dehmelt <i>et al.</i> 2006 [28]; Mazel <i>et al.</i> 2014 [29]
Chicken	Dorsal root ganglion neurons (primary culture)	EGFP- α -tubulin	Dynein	Grabham <i>et al.</i> 2007 [30]
Chicken	Dorsal root ganglion neurons (primary culture)	Docked mitochondria as fiducial markers for bulk cytoskeleton movements	Dynein	Roossien <i>et al.</i> 2014 [31]
<i>Drosophila melanogaster</i>	Larval brain neurons (primary culture) and S2 cells (tissue culture)	GFP-CAMSAP (minus-end microtubule marker) and EB1-GFP (plus-end microtubule marker)	Dynein	Del Castillo <i>et al.</i> 2015 [14]

hippocampal neurons describing shorter axons after kinesin-1 knockdown [19]. Furthermore, kinesin-dependent microtubule sliding is also required for neurite membrane extension during axon regeneration after injury in culture [16]. In a recently published study from our group, we were able to detect kinesin-dependent microtubule sliding in the growing sensory neurons underneath pupal bristles *in vivo* [6], further indicating that microtubule sliding is important for neuronal development.

Interestingly, another motor that can slide microtubules, cytoplasmic dynein, is dispensable for neurite outgrowth and axonal regeneration [8,14–16,20] (see more on the role of dynein in the section of 'Interplay between kinesin-1 and dynein in microtubule organization'). To experimentally test the kinesin-driven microtubule sliding mechanism presented in Figure 1, we generated a mutant form of kinesin that decreases its microtubule-sliding ability without affecting its cargo-transporting function. We mutated four highly conserved amino residues at the C-terminal microtubule binding site (914-RKRYQ-918 to 914-AAAYA-918) based on a human kinesin-1 (KIF5B) tail mutant that showed a 20-fold decrease in microtubule binding *in vitro* [4], creating Khc^{mutA}. Khc^{mutA} significantly reduced kinesin-1 tail microtubule binding and thus kinesin-driven microtubule sliding in *Drosophila* S2 cells, while conventional kinesin-driven

cargo transport was unaffected [6]. This sliding-deficient mutant allows us to experimentally dissociate the defects caused by impaired microtubule sliding from the defects caused by the reduced cargo transport. Based on the S2 work, we generated a genomic knock-in *Drosophila* line carrying this tail mutant, *Khc^{mutA}* [6]. Remarkably, *Khc^{mutA}* displayed reduced in kinesin-driven microtubule sliding in cultured neurons and neurons *in vivo*. Furthermore, *Khc^{mutA}* flies showed major defects of neurite outgrowth in culture, and of axonal and dendritic patterning *in vivo* [6]. These defects in the nervous system lead to decreased locomotion and increased lethality, which indicates that kinesin-driven microtubule sliding is an essential developmental process in addition to the well-recognized kinesin-driven cargo transport.

A similar tail mutant in the C-terminal microtubule binding site was generated in *C. elegans* by the group of Shen [21] who showed that microtubule polarity in dendrites, but not kinesin-driven mitochondria transport, is affected in this mutant. Shen and colleagues proposed that kinesin-1 slides plus-ends out microtubules out of the dendrites (consistent with the model presented in Figure 2A).

Both the motor domain and the ATP-independent microtubule binding site at the C terminus of the heavy chain are highly evolutionary conserved [5,6,22] (Figure 1). Therefore, we propose that microtubule sliding can serve as a universal and essential mechanism driving reorganization of the microtubule networks. In agreement with this idea, we have observed microtubule sliding in mammalian cells [8]; furthermore, human kinesin-1 heavy chain (KIF5B) fully rescues microtubule sliding in *Drosophila* S2 cells [6]. Because neurons are the most polarized cell type in animals, microtubule sliding is essential for neurons by providing the mechanical force for cell shape changes during neuronal differentiation and regeneration.

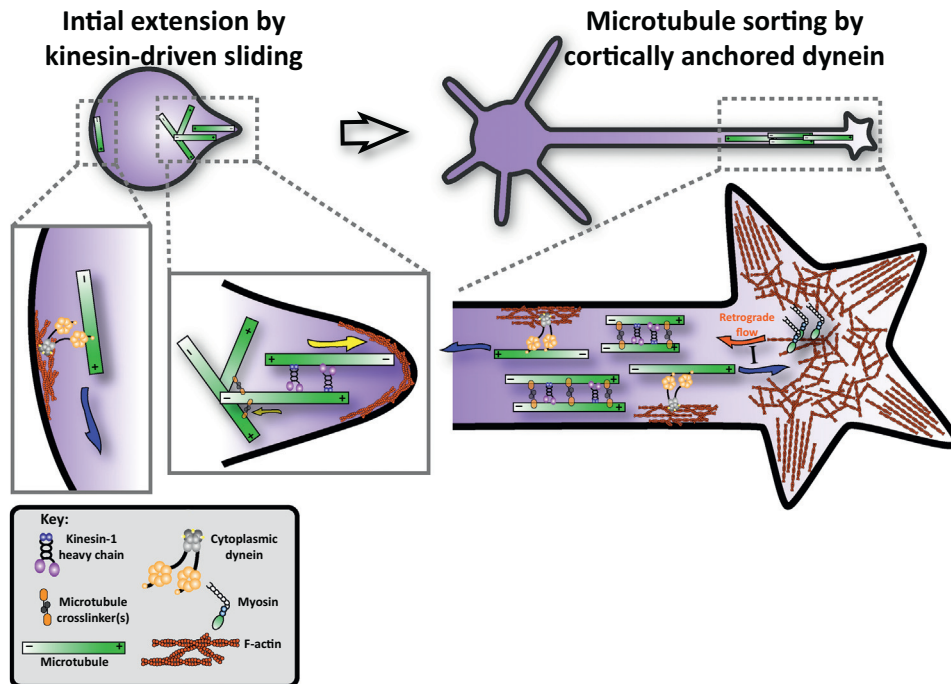
Interplay Between Kinesin-1 and Dynein in Microtubule Organization

In addition to the plus-end motor kinesin-1, minus-end motor cytoplasmic dynein has also been implicated as a motor for microtubule movement in neurons and tissue culture cells.

Using cultured rat sympathetic neurons, the Baas and colleagues have shown that dynein is important for the transport of microtubule fragments released from centrosomes, and for the anterograde transport of microtubules in axons [23–27]. Dehmelt and colleagues demonstrated that dynein pushes bundled microtubules outwards in Neuro2A neuroblastoma cell and primary hippocampal neurons, and that this movement is important for neurite initiation [28]. Directional movement and bending of microtubules is associated with and dependent on cortical dynein in COS7 cells and Neuro2A cells [29]. The group of Vallee showed that dynein is required for microtubule advance into the growth cone during axonal outgrowth in both chicken DRG neurons and rat hippocampal neurons [30]. This is further supported by a study from Miller and colleagues showing that *en masse* microtubule anterograde translocation was abolished upon dynein inhibition in chicken DRG neurons [31].

The popular model of dynein-driven microtubule movement is that the dynein complex binds to an immobilized stable scaffold, such as cortical F-actin, and pushes microtubules in the cytoplasm [28,31,32] (Figure 4). Alternatively, dynein can potentially bind to two microtubules simultaneously, using its two motor domains, and slide them apart. Such a possibility was demonstrated *in vitro* [33], although it is unclear whether this takes place in cells.

Interestingly, kinesin-driven and dynein-driven sliding models are not mutually exclusive. A recent study from our group showed that kinesin-1 and dynein activities are coordinated and function at different developmental stages to ensure normal neurite outgrowth and correct organization of microtubules in the axons of *Drosophila* larval brain neurons [34]. In this study we showed that, at early stages of initial neurite outgrowth, kinesin-1-driven microtubule sliding



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Figure 4. A Model of Microtubule Motors Driving Microtubule Sliding at Different Stages. Kinesin-1 drives the sliding apart of antiparallel microtubules in early-stage neurons, inducing the formation of initial neurites in spherical-shaped young neurons (left panel). At later stages, cytoplasmic dynein anchors at cell cortex by interacting with F-actin, and sorts out microtubules in the wrong orientation (minus-ends out), and slides microtubules of right orientation (plus-ends out) towards the growth cone to counteract actin retrograde flow driven by myosins (right panel). Both kinesin-driven and dynein-driven microtubule sliding can be regulated by microtubule crosslinker(s).

breaks the spherical symmetry of undifferentiated neurons, and drives initial neurite extensions, with minus-ends of microtubules pushing out the plasma membrane at the tips of the processes. This sliding can be favored in the anterograde direction because microtubules tangled or crosslinked in the cell body restrict retrograde sliding and provide the mechanical support for initial neurite extension (Figure 4, left panel). Later, cytoplasmic dynein becomes essential for microtubule sorting, establishing the uniform (plus-ends out) orientation of microtubules in the axons by sliding 'minus-end-out' microtubules towards the cell body (Figure 4, right panel). It can potentially also participate in outgrowth by moving plus-end out microtubules towards the tips, as demonstrated by other laboratories [23–26,30,31]. Dynein sorting function requires its recruitment to actin cortex because depolymerization of F-actin phenocopied dynein inhibition, while artificial recruitment of dynein to the plasma membrane bypassed the requirement for F-actin [34]. This is consistent with a recent report from the group of Hoogenraad showing that initially the neurites of rat hippocampal and cortical neurons contain microtubules of mixed polarity; in later stages the mixed orientation in axons changes to the uniform (plus-ends-out), while the mixed orientation in dendrites is preserved [35]. This suggests that during polarization the sorting machinery removes minus-end-out microtubules only from axons. In addition to its function of sorting minus-ends-out microtubules, dynein may mediate the forward movement of plus-ends-out microtubules, antagonizing myosin-mediated actin retrograde flow and neurite retraction [26,30,31] (Figure 4, right panel).

What triggers this transition from kinesin-driven to dynein-driven sliding is not unknown. The easiest explanation is that the transition is simply triggered by the changing geometry of the

developing neuron. Cortical dynein alone in a spherical cell cannot initiate process formation because by definition it would only move microtubules parallel to the cell surface, while kinesin-1 slides microtubules to form initial neurite extension by pushing against the plasma membrane (Figure 4, left panel). After the initial process is formed, dynein tethered to the cortex of the process started to drive both outgrowth, by pushing the plus-ends-out microtubules forward, and sorting by removing the minus-ends-out microtubules because the force produced by cortical dynein in the process will be directed parallel to the axis of the axon. At the same time, the ability of kinesin-1 to slide microtubules would be constrained because kinesin-1 cannot slide parallel microtubules in the axons, and instead bundles them (Figure 2B and Figure 4, right panel). Thus, simple geometry can explain the shift from kinesin-driven to dynein-driven neurite extension. In addition to cell geometry, biochemical regulation can also contribute to the switch between kinesin-1 and dynein sliding activity (see below).

Developmental Regulation of Microtubule Sliding

Because microtubule sliding is essential for axon elongation and regeneration, it is not surprising that sliding is developmentally regulated. Microtubule sliding is only active in young growing neurons, and becomes undetectable as they mature [15,16]. Downregulation of microtubule sliding is not caused by inhibition of kinesin-1 because kinesin-1-dependent organelle transport is still active in mature neurons [9,15,16,36]. We demonstrated that a 'mitotic' kinesin, Pavarotti (a kinesin-6 family member known in other organisms as MKLP1, CHO1, or KIF20A), is a potent negative regulator of kinesin-1-driven sliding of cytoplasmic microtubules [17], most likely by crosslinking antiparallel microtubules in neurons. Because Pavarotti contains multiple nuclear localization signals (NLS) at its C-terminal tail, and is mainly localized in the nucleus in interphase cells [37,38], it is likely that translocation of Pavarotti from the nucleus to the cytoplasm causes inhibition of microtubule sliding. This speculation is supported by the fact that ectopic expression of NLS-mutated Pavarotti resulted in dramatic inhibition of microtubule sliding and axon growth [17].

Microtubule sliding is downregulated in fully grown mature neurons; however, sliding can be reactivated by axonal injury. Reactivation depends on the formation of antiparallel microtubules near the injury site, and is essential for driving initial axonal regeneration [16]. It will be interesting to see whether suppression of kinesin-6 could stimulate axonal regeneration and, if so, whether it can be potentially used for treatment of neurodegenerative diseases as well as spinal cord and brain injuries.

Another example of developmental regulation of microtubule sliding is observed during mouse myogenesis. Microtubule movement has been demonstrated in mouse myoblast C2C12 cells during myogenesis and differentiation [39]. Straube and colleagues demonstrated that a novel isoform of microtubule-associated protein MAP4, oMAP4, functions as a microtubule cross-linker that inhibits microtubule sliding during myogenic differentiation [39]. The ability of oMAP4 to inhibit sliding is mostly regulated by its expression level: expression is upregulated during myogenic differentiation, when microtubule sliding is greatly reduced [39]. Thus, although microtubule crosslinking could serve as a general braking mechanism, the regulators themselves could be cell type-specific (oMAP4 during myogenic differentiation versus kinesin-6 in neurons).

Moreover, developmental downregulation of microtubule sliding observed in neurons [15–17] may help to resolve the longstanding debate about whether tubulin is transported as a polymer or as subunits in the axons [40,41]. Microtubule movement along the axons has been reported in *Xenopus* embryonic neurons [42,43] and rat sympathetic neurons [23,44], but no microtubule movement could be detected in rat neuron-like PC-12 cells [45], chicken DRG neurons [46], or mouse sensory neuron [47]. It is interesting to note that neurons in which microtubule

movement was reported are characterized by fast neurite outgrowth ($>25 \mu\text{m/h}$) [15,42,48–50]. As we reported, microtubule movement rate is highly correlated with neurite growth rate [15,17]. Therefore, the discrepancies in the previous studies are likely explained by different neurite outgrowth rates at different development stages: the high level of microtubule sliding is only detected during fast outgrowth, whereas during slow neurite outgrowth microtubule sliding becomes undetectable and/or easily obscured by microtubule dynamics.

The Role of F-actin in the Growth Cone

Despite the fact that F-actin is at the extreme periphery of the leading edge of the growth cone, numerous studies have shown that it is dispensable for neurite outgrowth; instead, disruption of F-actin leads to faster neurite outgrowth, probably by allowing microtubules penetrate into the growth cone to drive the extension [18,51–57]. Consistent with these findings, we showed that both fragmentation of F-actin by cytochalasin D and its total depolymerization by latrunculin B stimulate neurite outgrowth in *Drosophila* embryonic neurons. In the absence of F-actin at the neurite tip, sliding microtubules push directly against the cell membrane to make membrane protrusions [15,17]. In addition, it has been shown that local disruption of F-actin (cytochalasin D and latrunculin B) in rat hippocampal neurons favors axonal specification, and global disruption of F-actin leads to the formation of multiple long axons [18]. Disruption of F-actin also caused defects in axon targeting and branching [54,55]. Thus, F-actin in the growth cone plays an important role in outgrowth control and growth cone steering by allowing microtubule penetration in part of the growth cone towards local guidance cues [58,59]. Uncoupling of microtubules and F-actin in the growth cone, such as in mutants of the microtubule and actin crosslinker, spectraplakins/short stop, led to mistargeting of axons *in vivo* [60,61].

Intriguingly, a recent paper from the laboratory of Svitkina [62] questioned these studies, and claimed that F-actin polymerization drives neurite extension. We cannot exclude the possibility that, at particular stages of slow axonal outgrowth, actin filaments do provide some driving force for axonal and/or dendritic outgrowth or steering. However, Chia *et al.* [62] did not directly show that actin polymerization indeed drives neurite extension. Instead, they showed that after treatment with cytochalasin D or latrunculin B some F-actin remains as a cluster ('actin bulb') at the collapsed growth cone of some growing neurites. They argued that actin polymerization is required for neurite extension. In fact, cytochalasin D binds to growing ends of F-actin and prevent the addition of actin monomers (G-actin) [63], while latrunculin B binds to G-actin and prevents F-actin assembly [64], which means that both drugs effectively inhibit actin dynamics. The remnants of actin staining at the neurite tip after drug treatment suggested that these actin filaments are stable, which, together with the bulk of other published data [15,17,18,51–57], argues against their conclusion that actin polymerization drives neurite extension. Furthermore, their interpretation failed to explain why actin-specific inhibitors increase rather than decrease the rate of axon outgrowth.

Cargo Riding on Moving Microtubules: Another Form of Hitchhiking

Cargo transport and microtubule sliding are two main and conserved functions of kinesin-1. It is possible that some cargoes can simply ride on sliding microtubules and be transported within the cells along microtubules. This hypothesis was first proposed based on the observations of organelles moving with sliding microtubules in the amoeba *Reticulomyxa* [65–67]. More recently, an observation of concerted movement of peroxisomes in *Drosophila* S2 cells led to the speculation that they are attached to a moving microtubule [12]. Our recent study of *Drosophila* ooplasmic streaming revisited this hitchhiking hypothesis. We showed that kinesin-driven microtubule sliding is required for cytoplasmic streaming in *Drosophila* oocytes [22]. Unlike sliding in neurons, microtubule sliding in oocytes occurs between the cortex-anchored and free microtubules. This sliding generates unidirectional movement of cytoplasm. Inhibition of sliding by a kinesin mutation (*Khc^{mutA}*, see section 'Kinesin-1-driven microtubule sliding in

neurons and non-neuronal cells') resulted in a more diffuse pattern of posterior determinants in the oocyte. We propose that cytoplasmic streaming refines the posterior determination by bringing determinants that are attached to moving microtubules to the posterior pole for proper anchorage [22]. This is consistent with previous studies suggesting that streaming transports polarity determinants in the oocytes [68–70]. This 'ride-on-microtubules' mode of transport can serve as a quick and efficient way of redistributing organelles and mRNAs/proteins in parallel with microtubule reorganization, especially in big cells such as oocytes. More validation and future studies in other cell types and higher organisms are needed to further test this hypothesis.

Concluding Remarks and Future Perspectives

With the recent studies from multiple groups, interphase microtubule sliding emerges as a novel driving force for neurite outgrowth and microtubule reorganization. Kinesin-1 and cytoplasmic dynein both play important roles in driving microtubule movements, and in defining cell shape and microtubule organization/orientation. Furthermore, new studies reveal that motor-driven microtubule sliding is tightly controlled by developmental mechanisms. The new studies provide insights into the mechanism and regulation of this microtubule movement in neurons and other cells, and generate many more interesting questions: how do kinesin-1 and dynein switch their activities between microtubule sliding and cargo transport? How is dynein anchored to the cortex to slide microtubules in the axons? Is the activity of dynein-sorting mechanism differently regulated in axons versus dendrites? What regulates the activity of crosslinkers that inhibit microtubule sliding in neurons and other cell types? More studies will be necessary to address these important issues (see Outstanding Questions).

Outstanding Questions

Is there a specific mechanism that switches motors from the cargo-transporting to a microtubule-transporting mode?

How are actin dynamics at the growth cone coupled to microtubule sliding?

How is dynein anchored to the cortex of neurites?

Does differential regulation and/or anchoring of dynein contribute to differentiation of axons and dendrites?

Is microtubule hitchhiking a major mode of cargo transport in any cell type? What types of cargoes are translocated by this mechanism?

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References

- Hirokawa, N. *et al.* (2009) Kinesin superfamily motor proteins and intracellular transport. *Nat. Rev. Mol. Cell Biol.* 10, 682–696
- Rice, S. *et al.* (1999) A structural change in the kinesin motor protein that drives motility. *Nature* 402, 778–784
- Hackney, D.D. and Stock, M.F. (2000) Kinesin's IAK tail domain inhibits initial microtubule-stimulated ADP release. *Nat. Cell Biol.* 2, 257–260
- Seeger, M.A. and Rice, S.E. (2010) Microtubule-associated protein-like binding of the kinesin-1 tail to microtubules. *J. Biol. Chem.* 285, 8155–8162
- Navone, F. *et al.* (1992) Cloning and expression of a human kinesin heavy chain gene: interaction of the COOH-terminal domain with cytoplasmic microtubules in transfected CV-1 cells. *J. Cell Biol.* 117, 1263–1275
- Winding, M. *et al.* (2016) Role of kinesin-1-based microtubule sliding in *Drosophila* nervous system development. *Proc. Natl. Acad. Sci. U. S. A.* 113, E4985–4994
- Straube, A. *et al.* (2006) Conventional kinesin mediates microtubule-microtubule interactions in vivo. *Mol. Biol. Cell* 17, 907–916
- Jolly, A.L. *et al.* (2010) Kinesin-1 heavy chain mediates microtubule sliding to drive changes in cell shape. *Proc. Natl. Acad. Sci. U. S. A.* 107, 12151–12156
- Barlan, K. *et al.* (2013) The microtubule-binding protein ensconsin is an essential cofactor of kinesin-1. *Curr. Biol.* 23, 317–322
- Ling, S.C. *et al.* (2004) Transport of *Drosophila* fragile X mental retardation protein-containing ribonucleoprotein granules by kinesin-1 and cytoplasmic dynein. *Proc. Natl. Acad. Sci. U. S. A.* 101, 17428–17433
- Kim, H. *et al.* (2007) Microtubule binding by dynactin is required for microtubule organization but not cargo transport. *J. Cell Biol.* 176, 641–651
- Kulic, I.M. *et al.* (2008) The role of microtubule movement in bidirectional organelle transport. *Proc. Natl. Acad. Sci. U. S. A.* 105, 10011–10016
- Ally, S. *et al.* (2009) Opposite-polarity motors activate one another to trigger cargo transport in live cells. *J. Cell Biol.* 187, 1071–1082
- Del Castillo, U. *et al.* (2015) Interplay between kinesin-1 and cortical dynein during axonal outgrowth and microtubule organization in *Drosophila* neurons. *eLife* 4, e10140
- Lu, W. *et al.* (2013) Initial neurite outgrowth in *Drosophila* neurons is driven by kinesin-powered microtubule sliding. *Curr. Biol.* 23, 1018–1023
- Lu, W. *et al.* (2015) Kinesin-1-powered microtubule sliding initiates axonal regeneration in *Drosophila* cultured neurons. *Mol. Biol. Cell* 26, 1296–1307
- del Castillo, U. *et al.* (2015) Pavarotti/MKLP1 regulates microtubule sliding and neurite outgrowth in *Drosophila* neurons. *Curr. Biol.* 25, 200–205
- Bradke, F. and Dotti, C.G. (1999) The role of local actin instability in axon formation. *Science* 283, 1931–1934
- Ferreira, A. *et al.* (1992) Suppression of kinesin expression in cultured hippocampal neurons using antisense oligonucleotides. *J. Cell Biol.* 117, 595–606
- Beaven, R. *et al.* (2015) *Drosophila* CLIP-190 and mammalian CLIP-170 display reduced microtubule plus end association in the nervous system. *Mol. Biol. Cell* 26, 1491–1508
- Yan, J. *et al.* (2013) Kinesin-1 regulates dendrite microtubule polarity in *Caenorhabditis elegans*. *eLife* 2, e00133
- Lu, W. *et al.* (2016) Microtubule-microtubule sliding by kinesin-1 is essential for normal cytoplasmic streaming in *Drosophila* oocytes. *Proc. Natl. Acad. Sci. U. S. A.* 113, E4995–5004
- Ahmad, F.J. and Baas, P.W. (1995) Microtubules released from the neuronal centrosome are transported into the axon. *J. Cell Sci.* 108, 2761–2769
- Ahmad, F.J. *et al.* (1998) Cytoplasmic dynein and dynactin are required for the transport of microtubules into the axon. *J. Cell Biol.* 140, 391–401
- He, Y. *et al.* (2005) Role of cytoplasmic dynein in the axonal transport of microtubules and neurofilaments. *J. Cell Biol.* 168, 697–703
- Ahmad, F.J. *et al.* (2000) Motor proteins regulate force interactions between microtubules and microfilaments in the axon. *Nat. Cell Biol.* 2, 276–280

27. Rao, A.N. *et al.* (2016) Sliding of centrosome-unattached microtubules defines key features of neuronal phenotype. *J. Cell Biol.* 213, 329–341
28. Dehmelt, L. *et al.* (2006) A microtubule-based, dynein-dependent force induces local cell protrusions: implications for neurite initiation. *Brain Cell Biol.* 35, 39–56
29. Mazel, T. *et al.* (2014) Direct observation of microtubule pushing by cortical dynein in living cells. *Mol. Biol. Cell* 25, 95–106
30. Grabham, P.W. *et al.* (2007) Cytoplasmic dynein and LIS1 are required for microtubule advance during growth cone remodeling and fast axonal outgrowth. *J. Neurosci.* 27, 5823–5834
31. Roossien, D.H. *et al.* (2014) Cytoplasmic dynein pushes the cytoskeletal meshwork forward during axonal elongation. *J. Cell Sci.* 127, 3593–3602
32. Muresan, V. and Muresan, Z. (2012) Unconventional functions of microtubule motors. *Arch. Biochem. Biophys.* 520, 17–29
33. Tanenbaum, M.E. *et al.* (2013) Cytoplasmic dynein crosslinks and slides anti-parallel microtubules using its two motor domains. *eLife* 2, e00943
34. Del Castillo, U. *et al.* (2015) Interplay between kinesin-1 and cortical dynein during axonal outgrowth and microtubule organization in neurons. *eLife* 4
35. Yau, K.W. *et al.* (2016) Dendrites in vitro and in vivo contain microtubules of opposite polarity and axon formation correlates with uniform plus-end-out microtubule orientation. *J. Neurosci.* 36, 1071–1085
36. Lu, W. *et al.* (2013) Organelle transport in cultured *Drosophila* cells: s2 cell line and primary neurons. *J. Visualized Exp.* 50838
37. Minestrini, G. *et al.* (2002) Domains of the Pavarotti kinesin-like protein that direct its subcellular distribution: effects of mislocalisation on the tubulin and actin cytoskeleton during *Drosophila* oogenesis. *J. Cell Sci.* 115, 725–736
38. Minestrini, G. *et al.* (2003) Localization of Pavarotti-KLP in living *Drosophila* embryos suggests roles in reorganizing the cortical cytoskeleton during the mitotic cycle. *Mol. Biol. Cell* 14, 4028–4038
39. Mogessie, B. *et al.* (2015) A novel isoform of MAP4 organises the paraxial microtubule array required for muscle cell differentiation. *eLife* 4, e05697
40. Baas, P.W. and Brown, A. (1997) Slow axonal transport: the polymer transport model. *Trends Cell Biol.* 7, 380–384
41. Hirokawa, N. *et al.* (1997) Slow axonal transport: the subunit transport model. *Trends Cell Biol.* 7, 384–388
42. Reinsch, S.S. *et al.* (1991) Microtubule polymer assembly and transport during axonal elongation. *J. Cell Biol.* 115, 365–379
43. Tanaka, E.M. and Kirschner, M.W. (1991) Microtubule behavior in the growth cones of living neurons during axon elongation. *J. Cell Biol.* 115, 345–363
44. Wang, L. and Brown, A. (2002) Rapid movement of microtubules in axons. *Curr. Biol.* 12, 1496–1501
45. Lim, S.S. *et al.* (1989) Progressive and spatially differentiated stability of microtubules in developing neuronal cells. *J. Cell Biol.* 109, 253–263
46. Lim, S.S. *et al.* (1990) A test of microtubule translocation during neurite elongation. *J. Cell Biol.* 111, 123–130
47. Okabe, S. and Hirokawa, N. (1992) Differential behavior of photo-activated microtubules in growing axons of mouse and frog neurons. *J. Cell Biol.* 117, 105–120
48. Slaughter, T. *et al.* (1997) Microtubule transport from the cell body into the axons of growing neurons. *J. Neurosci.* 17, 5807–5819
49. Yu, W.Q. and Baas, P.W. (1995) The growth of the axon is not dependent upon net microtubule assembly at its distal tip. *J. Neurosci.* 15, 6827–6833
50. Yu, W. *et al.* (1996) Microtubule transport and assembly during axon growth. *J. Cell Biol.* 133, 151–157
51. Marsh, L. and Letourneau, P.C. (1984) Growth of neurites without filopodial or lamellipodial activity in the presence of cytochalasin-B. *J. Cell Biol.* 99, 2041–2047
52. Forscher, P. and Smith, S.J. (1988) Actions of cytochalasins on the organization of actin filaments and microtubules in a neuronal growth cone. *J. Cell Biol.* 107, 1505–1516
53. Joshi, H.C. *et al.* (1985) Tension and compression in the cytoskeleton of PC 12 neurites. *J. Cell Biol.* 101, 697–705
54. Bentley, D. and Toroian-Raymond, A. (1986) Disoriented path-finding by pioneer neurone growth cones deprived of filopodia by cytochalasin treatment. *Nature* 323, 712–715
55. Dent, E.W. and Kalil, K. (2001) Axon branching requires interactions between dynamic microtubules and actin filaments. *J. Neurosci.* 21, 9757–9769
56. Ruthel, G. and Hollenbeck, P.J. (2000) Growth cones are not required for initial establishment of polarity or differential axon branch growth in cultured hippocampal neurons. *J. Neurosci.* 20, 2266–2274
57. Flynn, K.C. *et al.* (2012) ADF/cofilin-mediated actin retrograde flow directs neurite formation in the developing brain. *Neuron* 76, 1091–1107
58. Geraldo, S. and Gordon-Weeks, P.R. (2009) Cytoskeletal dynamics in growth-cone steering. *J. Cell Sci.* 122, 3595–3604
59. Kahn, O.I. and Baas, P.W. (2016) Microtubules and growth cones: motors drive the turn. *Trends Neurosci.* 39, 433–440
60. Lee, S. *et al.* (2000) *short stop* is allelic to *kakapo*, and encodes rod-like cytoskeletal-associated proteins required for axon extension. *J. Neurosci.* 20, 1096–1108
61. Lee, S. and Kolodziej, P.A. (2002) Short Stop provides an essential link between F-actin and microtubules during axon extension. *Development* 129, 1195–1204
62. Chia, J.X. *et al.* (2016) Neurite outgrowth is driven by actin polymerization even in the presence of actin polymerization inhibitors. *Mol. Biol. Cell* 27, 3695–3704
63. Casella, J.F. *et al.* (1981) Cytochalasin D inhibits actin polymerization and induces depolymerization of actin filaments formed during platelet shape change. *Nature* 293, 302–305
64. Spector, I. *et al.* (1989) Latrunculin – novel marine macrolides that disrupt microfilament organization and affect cell-growth. 1. Comparison with cytochalasin-D. *Cell Motil. Cytoskeleton* 13, 127–144
65. Koonce, M.P. *et al.* (1987) Active sliding between cytoplasmic microtubules. *Nature* 328, 737–739
66. Euteneuer, U. *et al.* (1989) Microtubule bundles of *Reticulomyxa* networks are of uniform polarity. *Eur. J. Cell Biol.* 49, 373–376
67. Orokos, D.D. *et al.* (2000) Organelles are transported on sliding microtubules in *Reticulomyxa*. *Cell Motil. Cytoskeleton* 47, 296–306
68. Glotzer, J.B. *et al.* (1997) Cytoplasmic flows localize injected oskar RNA in *Drosophila* oocytes. *Curr. Biol.* 7, 326–337
69. Forrest, K.M. and Gavis, E.R. (2003) Live imaging of endogenous RNA reveals a diffusion and entrapment mechanism for nanos mRNA localization in *Drosophila*. *Curr. Biol.* 13, 1159–1168
70. Mische, S. *et al.* (2007) Direct observation of regulated ribonucleoprotein transport across the nurse cell/oocyte boundary. *Mol. Biol. Cell* 18, 2254–2263